

Week 2 – Introduction to Course

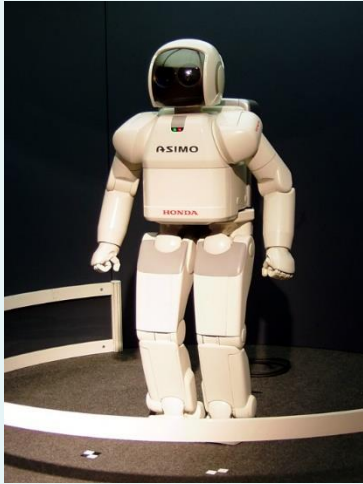
ELEC0129 Introduction to Robotics

Dr. Chow Yin Lai

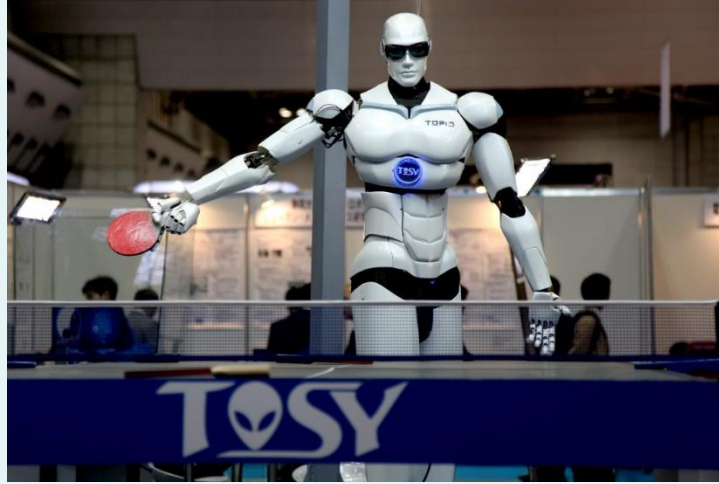
Email: uceecyl@ucl.ac.uk

Types of Robots – Appearance (1)

- Humanoid robots:



Honda ASIMO [1]



TOSY Ping Pong Playing Robot [2]



Atlas from Boston Dynamics [3]

Types of Robots – Appearance (2)

- Animal-Like Robots:



Sony AIBO Robot Dog [4]



Snake Robot [5]



Jessiko Swimming Robot [6]

Kangaroo Robot: <https://www.youtube.com/watch?v=mWiNIWk1Muw>

Types of Robots – Applications (1)

- Service robots:
 - Perform services useful to the well-being of humans.
 - Excludes manufacturing operations.



Robotic Vacuum
Cleaner [7]



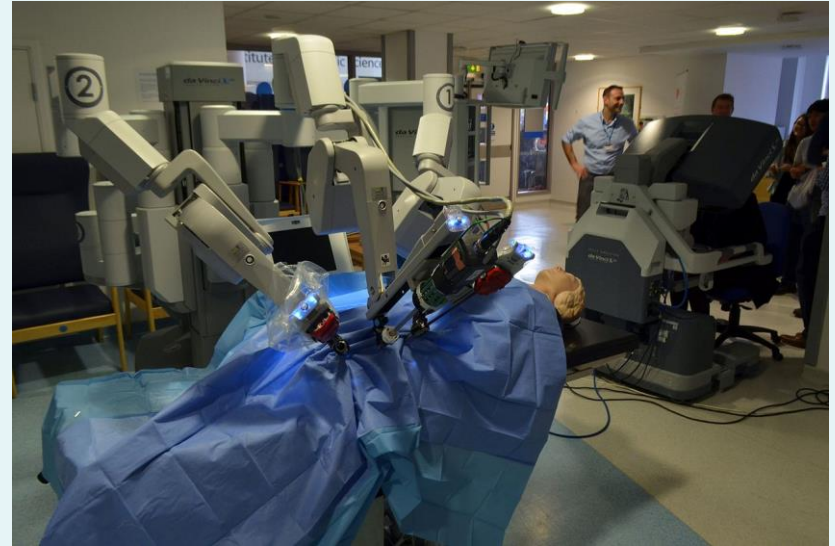
Robotic Waiters [8]



Care-O-Bot by
Fraunhofer [9]

Types of Robots – Applications (2)

- Surgical robots:
 - Allows doctors to perform complex procedures with more precision.
 - Usually used in minimally invasive surgery.



Da Vinci Robot [10]

Types of Robots – Applications (3)

- Mobile robots / Automated guided vehicles:
 - Moving of materials around manufacturing facility or warehouse.



Automated Forklift [11]



Amazon Warehouse Robots [12]

Types of Robots – Applications (4)

- Education robots:
 - Teaches language, singing etc.
 - Helps students with autism.



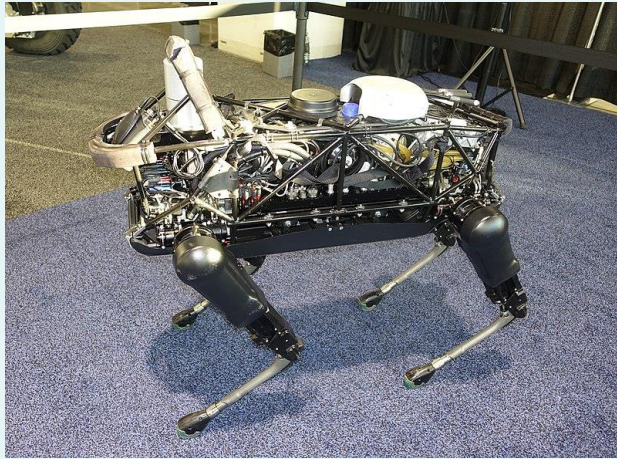
Keepon Robot [13]



NAO Robot [14]

Types of Robots – Applications (5)

- Military robots:



Big Dog [15]

<https://www.youtube.com/watch?v=cNZPRsrwumQ>



Robot checking for explosives [16]

Types of Robots – Applications (6)

- Industrial Robotics Manipulator:
 - An “automatically controlled, programmable, multipurpose manipulator programmable in three or more axes,
 - which may be either fixed in place or mobile for use in industrial automation applications.” (ISO 8373)



Industrial Robots [17]

Industrial Robotics Manipulator (1)

- In this course, we will focus on **Industrial Robotic Manipulators**.
- The fundamentals can also be applied to many other types of robots.
- To be further explored in research projects.



Industrial Robots on
Mobile Platform [18]

Industrial Robotics Manipulator (2)

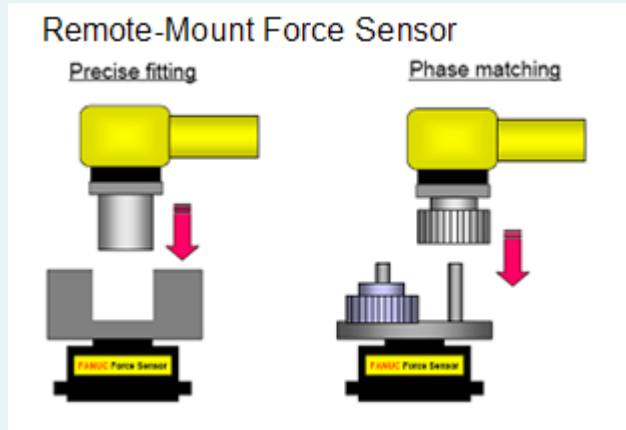
- Industrial robotic manipulators are widely used in:
 - Pick-and-Place.
 - E.g. Rapidly taking drink cartons from conveyor belt and placing them into box.
 - Machine tending.
 - E.g. Loading and unloading machines.
 - Welding.
 - Spray painting.
 - Dirty, Dull & Dangerous (3D) Tasks.



Welding Robots [19]

Industrial Robotics Manipulator (3)

- They are also increasingly being used for contact type operations which need a **sense of touch / force**:
 - Assembly (Peg in hole)
 - Polishing
 - Deburring



Assembly [20]



Robot Polishing [21]

Industrial Robotics Manipulator (4)

- Two main types:
 - Serial manipulators.
 - Parallel manipulators.



Serial robot [22]



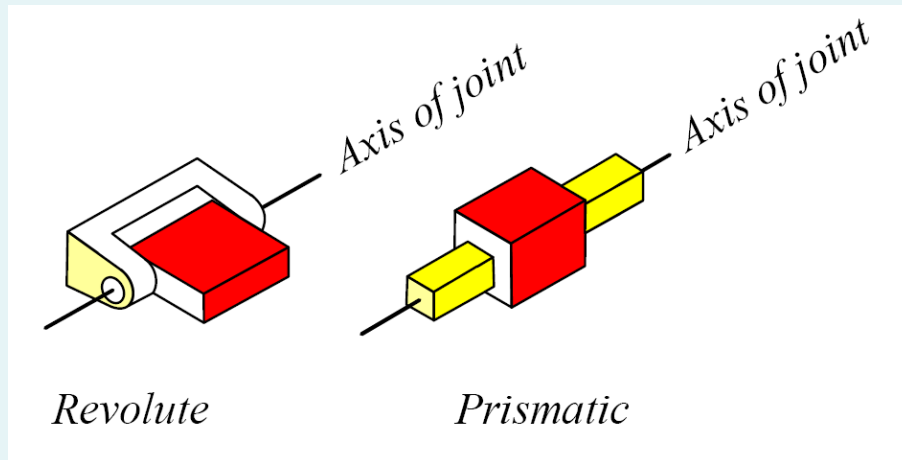
SCARA robot [23]



DELTA robot [24]

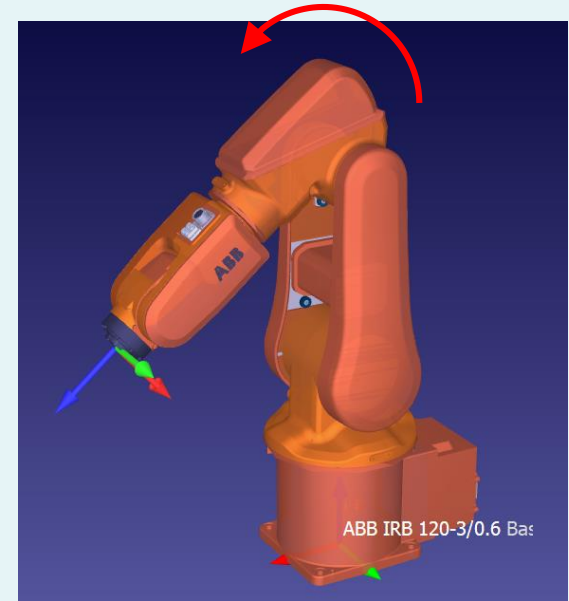
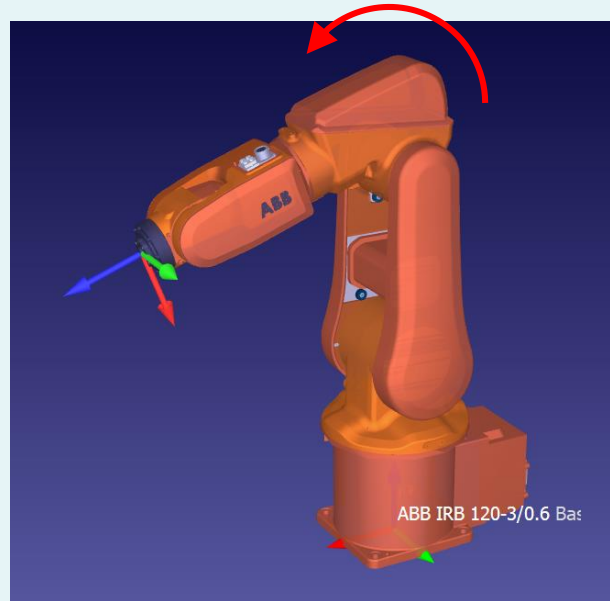
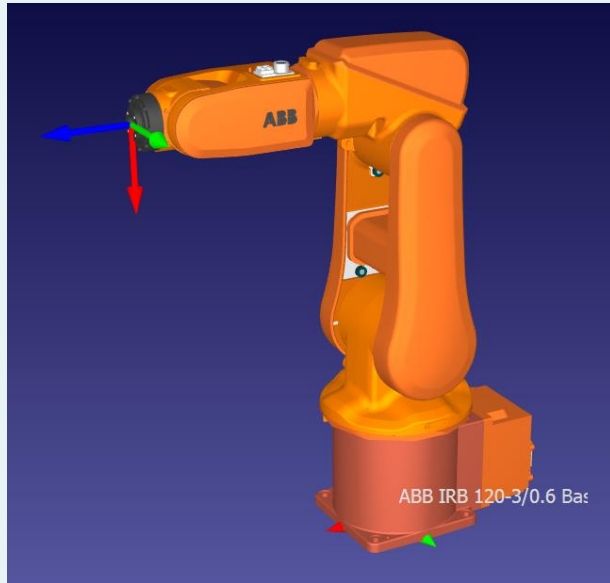
Terminologies (1)

- Industrial robots are mostly made by **links** connected by **joints**.
 - The joints could be **revolute (R)** or **prismatic (P)**.



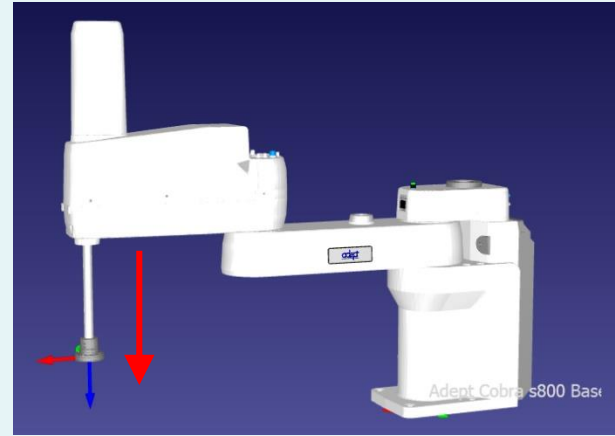
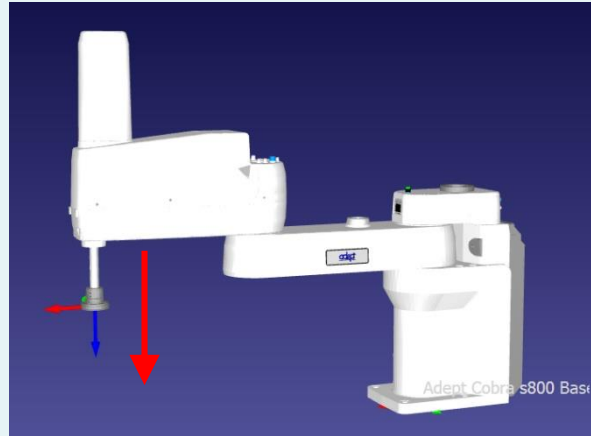
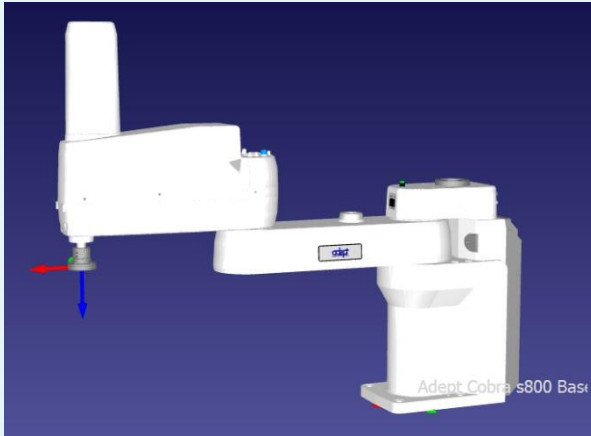
Terminologies (2)

- Revolute joints:



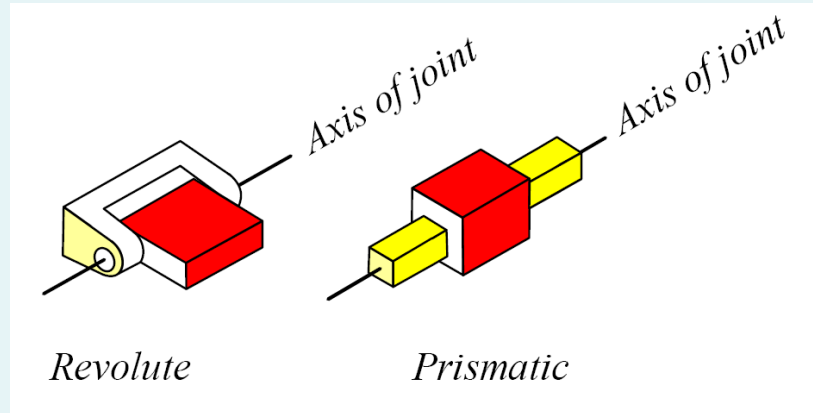
Terminologies (3)

- Prismatic joints:



Terminologies (4)

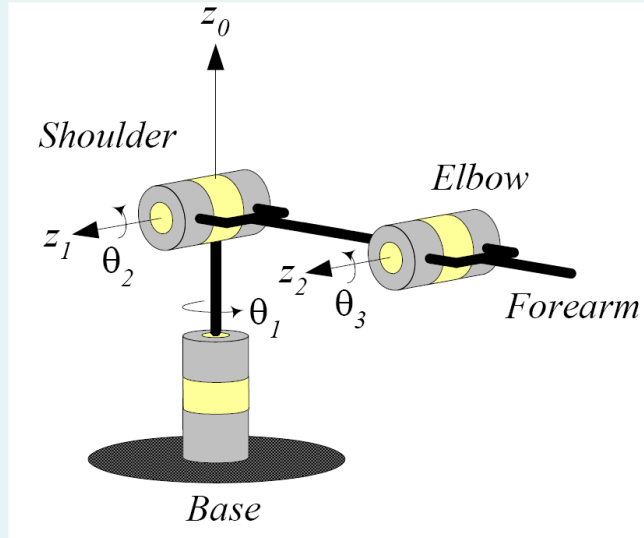
- Each joint allows only one **degree-of-freedom** (DOF).



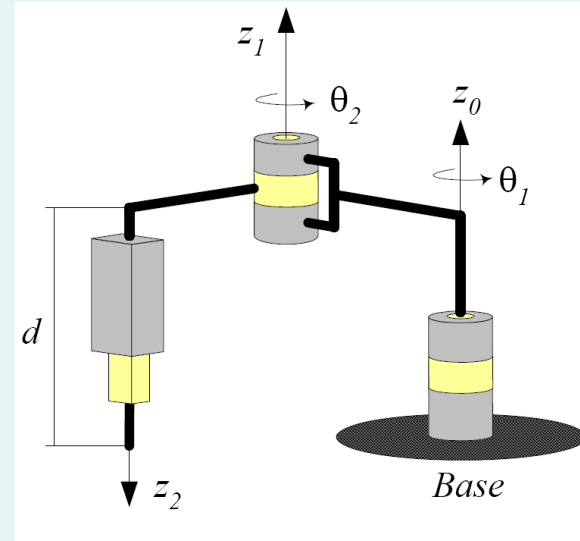
- Questions: What does DOF mean? How many DOFs does an unconstrained object have in free space? – See lecture video.

Terminologies (5)

- Examples of 3-DOF Manipulators (3 joints):



3R Robot

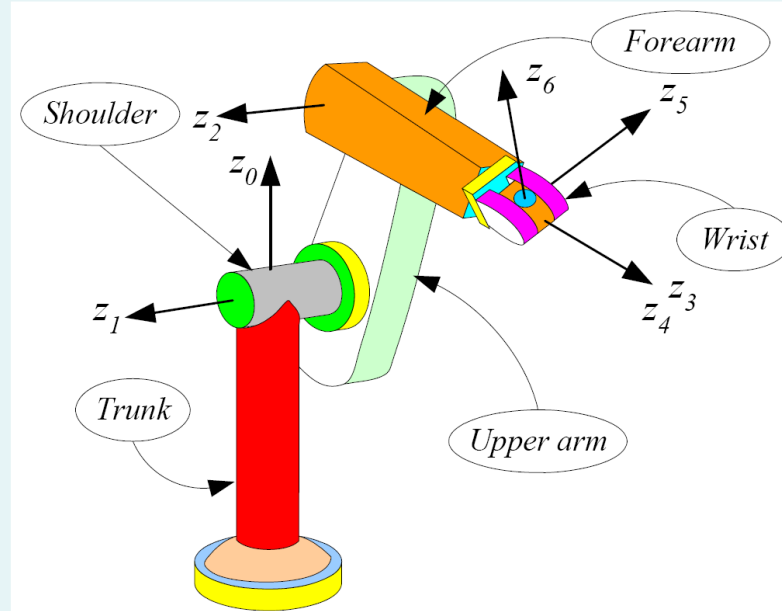


RRP Robot

- Question: Which DOFs do these robot have?

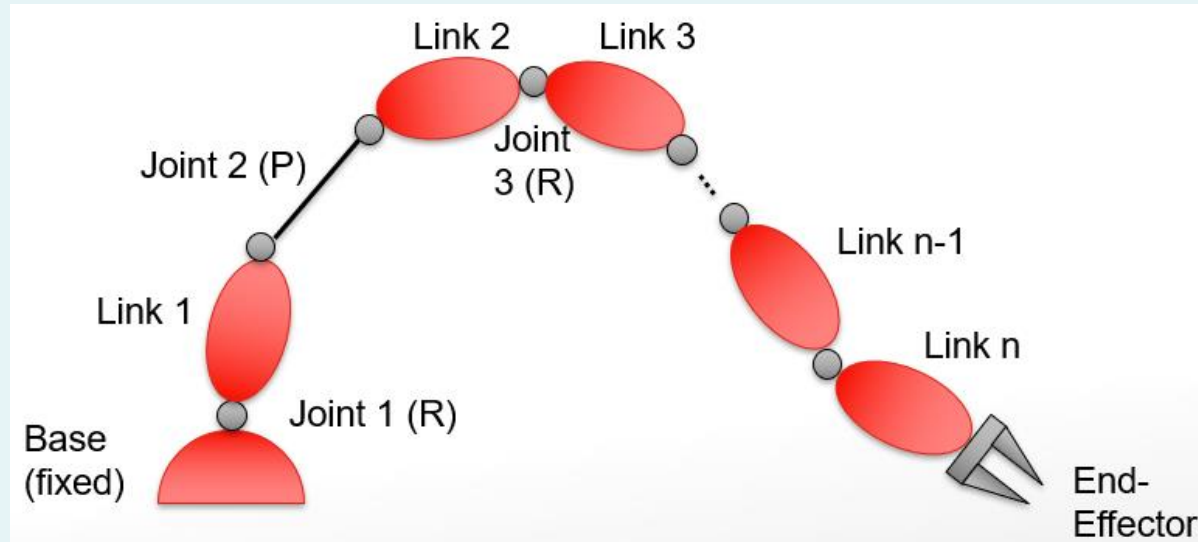
Terminologies (6)

- Example of a 6-DOF Manipulator (6 joints).



Terminologies (7)

- The links and joints are numbered from the robot base (0) to the tip (n).



Terminologies (8)

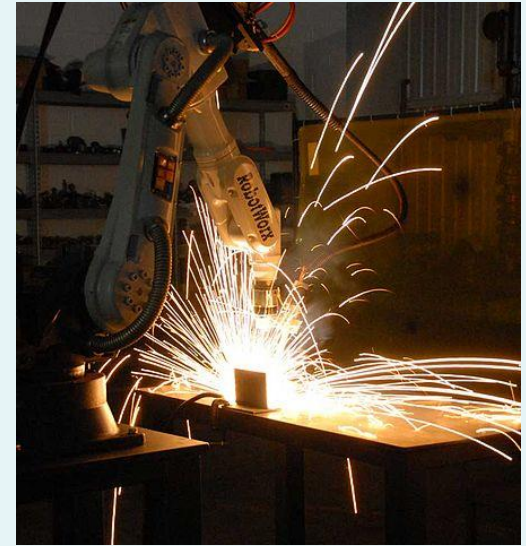
- End-Effectors:
 - Device attached to the end of a robot.
 - Interacts with the environment.



Vacuum Gripper [25]



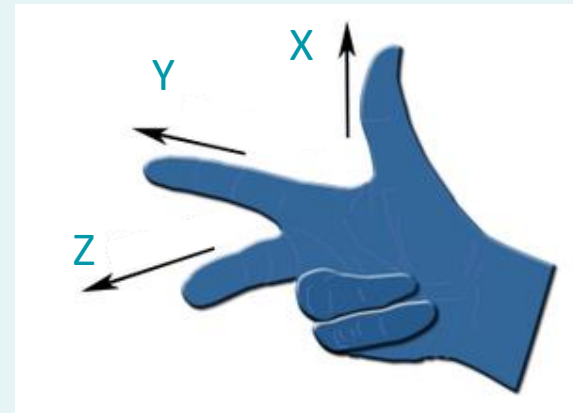
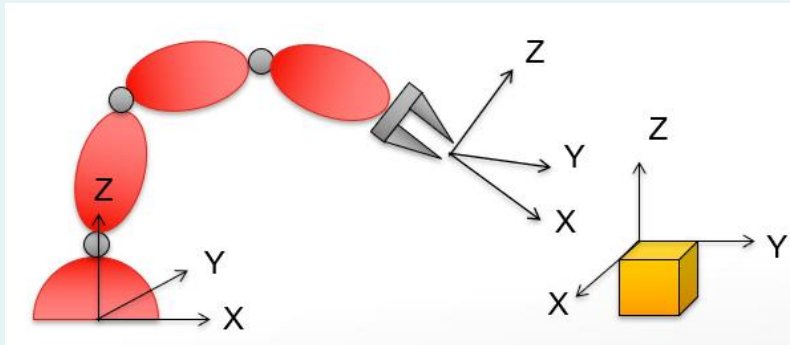
Pinching Gripper [26]



Welding Torch [27]

Terminologies (9)

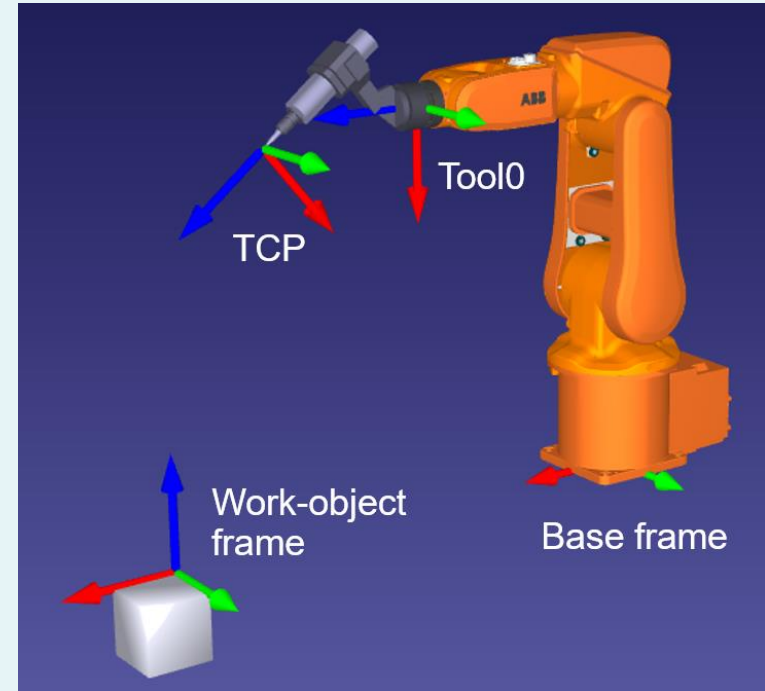
- Frames:
 - **Coordinate systems**, acting as reference for the description of position and orientation of objects.
 - **RGB colour code**: Red for X, Green for Y, Blue for Z.
 - Must follow **right hand rule**.



Right hand rule [28]

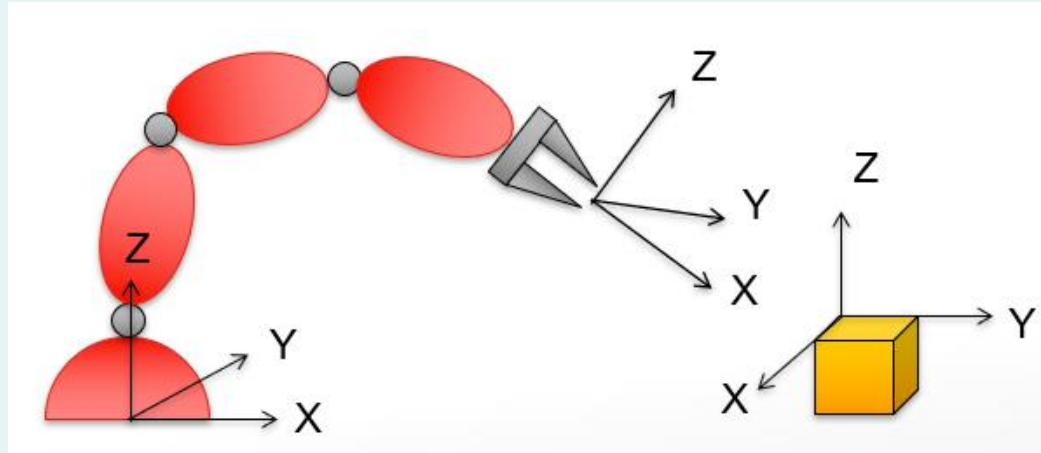
Terminologies (10)

- Special frames on robotic systems:
 - **Base frame**: Base of robot, fixed.
 - Work-object frame (**Wobj**): Attached to the object or work table.
 - **Tool0**: Tip of robot.
 - Tool Centre Point (**TCP**): Working point of end-effector.



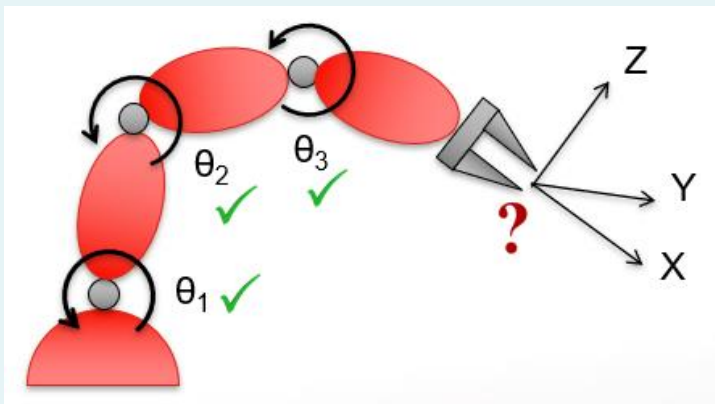
Topics Covered in this Course (1)

- Description of **position** and **orientation**.
 - In robotics, location of objects in 3-D space is of utmost importance.
 - Objects include parts, tools, end-effector, and the manipulator itself.
 - We will learn how to describe the position and orientation of the objects.



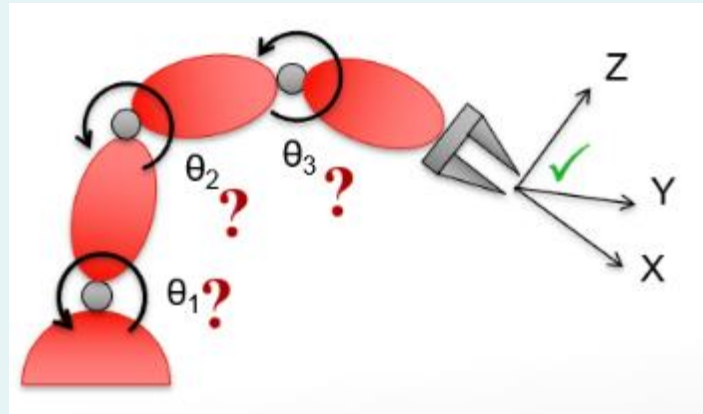
Topics Covered in this Course (2)

- Forward kinematics:
 - Given the **joint space** parameters (angles for revolute joints, or offsets for prismatic joints),
 - what is the position and orientation of the end-effector in **Cartesian space**?



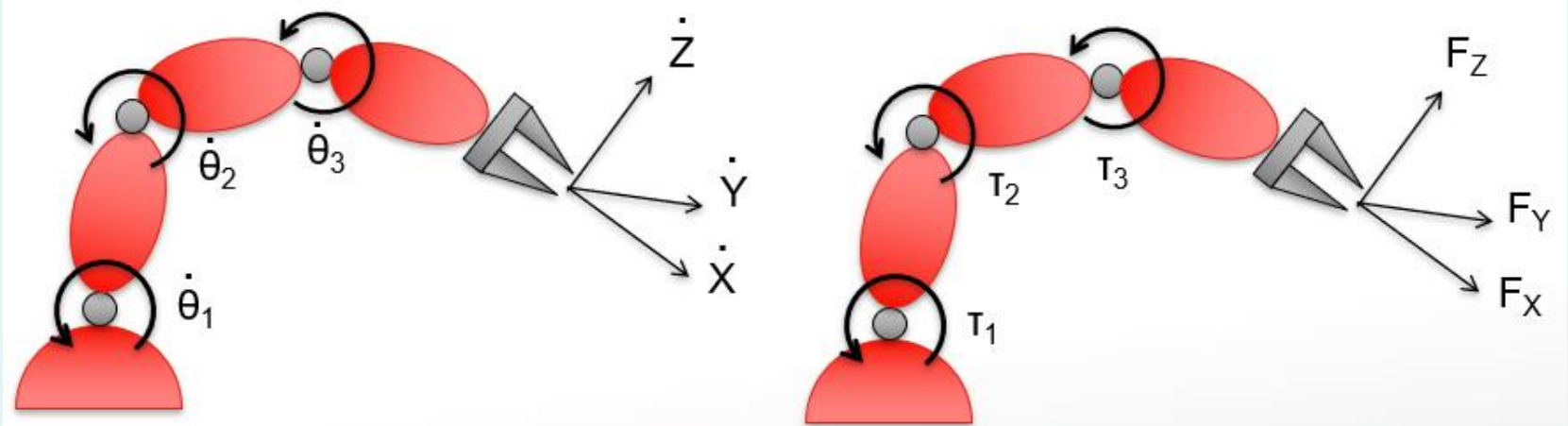
Topics Covered in this Course (3)

- Inverse kinematics:
 - Given the position and orientation of the end-effector in Cartesian space,
 - what are the parameters in joint space (joint angles or offsets)?



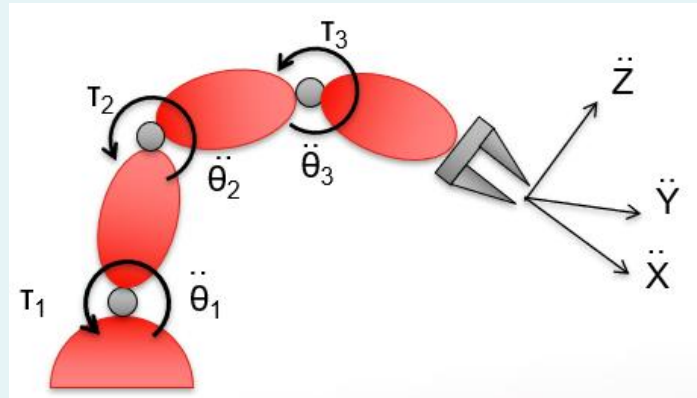
Topics Covered in this Course (4)

- **Jacobians:** Velocities and Static Forces
 - Relationship between **joint velocity** and in **Cartesian velocity**.
 - Relationship between the **joint torques** and the **static force** applied by end-effector onto a work surface.



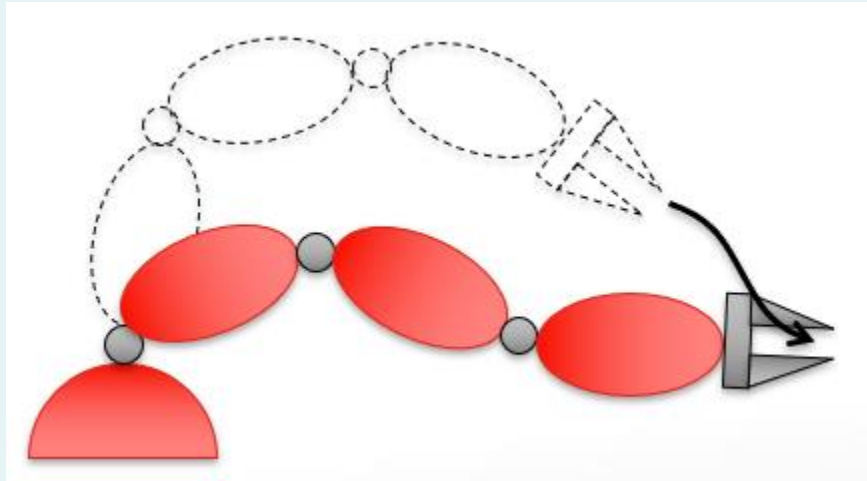
Topics Covered in this Course (5)

- Manipulator **Dynamics**:
 - How much **torque** is needed to **accelerate** the manipulator from rest to constant velocity, and then back to stop?
 - Dynamics also provide us a model (**equations of motions**) for simulation and control design purpose.



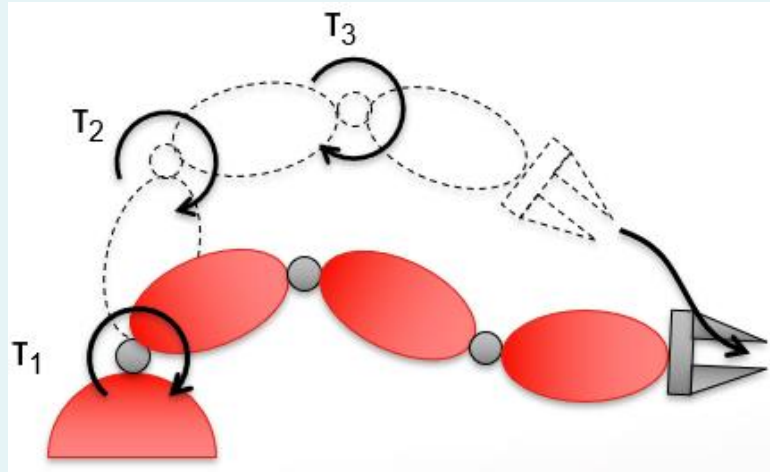
Topics Covered in this Course (6)

- Trajectory Planning
 - Design of a **time profile** for the manipulator to follow.



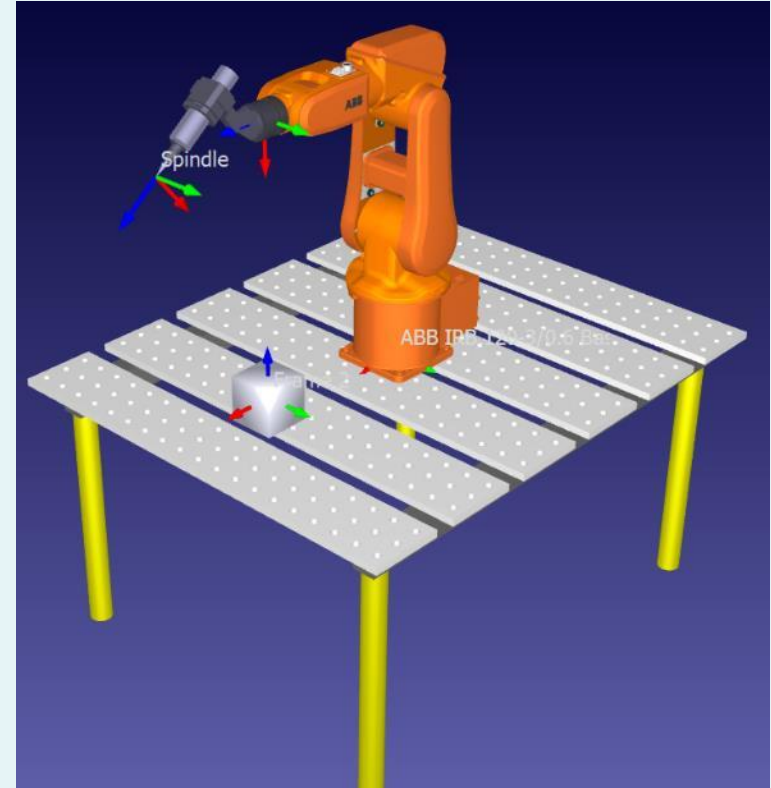
Topics Covered in this Course (7)

- **Control** of Robotics Manipulator.
 - Control the torque of the motors at the joints,
 - to execute the planned trajectory.



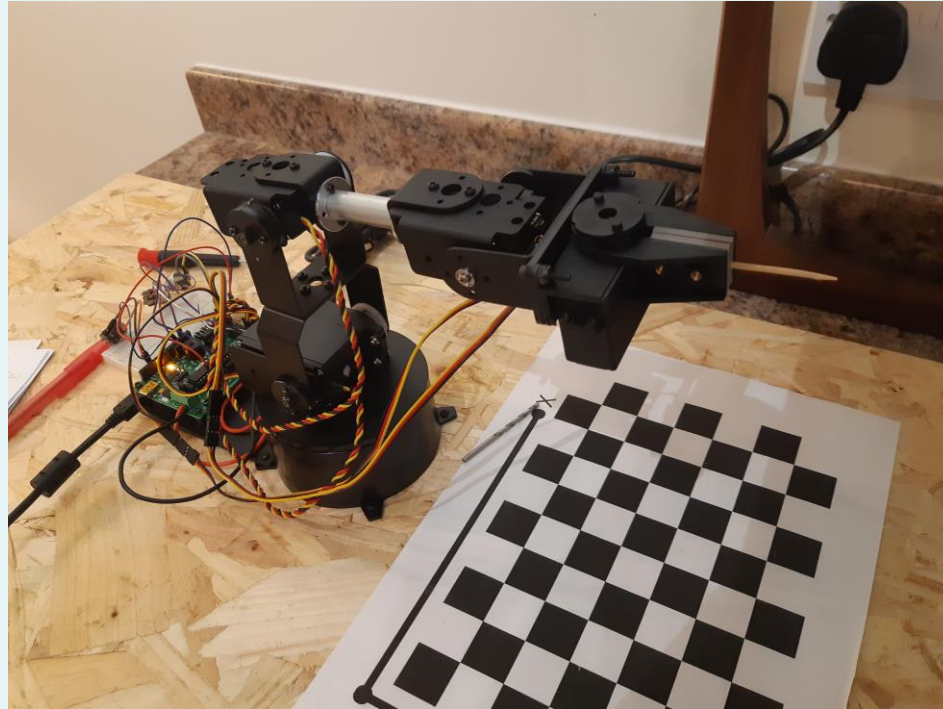
Topics Covered in this Course (8)

- Robotic **Offline Programming**.
 - Use offline programming software (RoboDK) to program industrial robots.



Topics Covered in this Course (9)

- You will also build your own robot to perform pick-and-place task!



Lectures

- Lectures will be **pre-recorded** and uploaded to Moodle latest by **Fridays of the previous week**.
 - E.g. Week 2 materials will be released latest by Friday of Week 1.
- Consistent and active attendance in **ALL** lectures will significantly help you obtain a good mark in the final exam.
 - The final exam is mainly focused on what was explained in the lectures.

Tutorials

- Tutorials will also be **pre-recorded** and uploaded to Moodle latest by **Fridays of the previous week**.
- Please try to **solve the questions yourselves** first, before watching the videos for solutions.
- This will help you find out if you understand the lecture materials.



[This Photo](#) by Unknown
Author is licensed under [CC BY](#)

Workshops

- Workshops will **run live** during the time-tabled slots.
- There will be **no recordings**, as you would follow the instruction sheets to complete the workshop.
 - The live session is for Q&A and support.
- The workshops will mostly be done in teams.



[This Photo](#) by Unknown Author is licensed under [CC BY](#)

Schedule

Legend:

A: MPEB 6.02 Teaching Lab

B: Roberts 7.04 PC Lab

Week	Recorded, uploaded by Friday of previous week	Workshop, Mondays 11am-1pm and/or Wednesdays 9am-11am	Notes
1	(Scenario Week)		
2	Lec: Intro; Spatial description	A: Workshop: Offline programming	Half class Mon, half class Wed
3	Lec & Tut: Spatial description	A: Workshop: Build robot	
4	Lec & Tut: Forward kinematics	A: Workshop: Forward kinematics	
5	Lec & Tut: Inverse kinematics	B: Workshop: Offline programming	Half class Mon, half class Wed
RW	(Reading Week)		
6	(Scenario Week)		
7	Lec & Tut: Jacobians	A: Workshop: Inverse kinematics	
8	Lec: Trajectory Planning	A: Workshop: Trajectory planning	
9	Lec & Tut: Dynamics	A: Workshop: Trajectory planning	
10	Lec & Tut: Control	A: Workshop: Pick-and-place demo	

Assignments

- **Robot offline programming** (Individual, 20%)
 - Start in week 2, due 5pm Friday of Reading Week.
- **Build a robot** (Group of Four, 40%)
 - Start in week 3.
 - Report (30%) on forward kinematics, inverse kinematics and trajectory planning, due 5pm Friday of Week 9.
 - Pick-and-place demonstration (10%), during workshop hours of Week 10.
- **Final exam** (Individual, 40%)
 - Closed-book exam.
 - Basic calculators allowed.



[This Photo](#) by Unknown Author is licensed under [CC BY-NC-ND](#)

Peer Assessment (1)

- Team members will rate each other based on agreed criteria.
- The individual score will then be calculated as follows:

Individual score = Group score x Individual rating



[This Photo](#) by Unknown
Author is licensed under
[CC BY-NC](#)

Peer Assessment (2)

- Submission of peer assessment is compulsory.
- If you do not submit this on time:
 - your individual rating will be capped at 70%.



[This Photo](#) by Unknown Author is licensed under [CC BY-NC-ND](#)

Communication

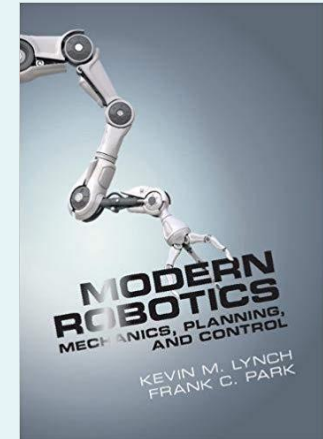
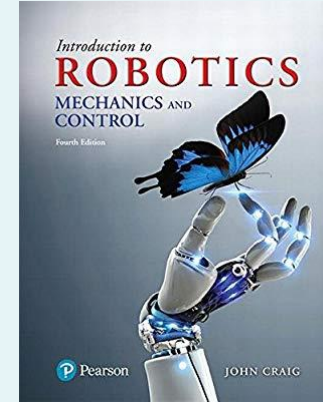
- If you have **questions**, please ask **during workshop sessions** or post in **Moodle Forum**.
 - This would benefit everyone in the class.
 - Questions in Moodle forum will be answered within **two working days**.
- **Office hour:**
 - Flexible.
 - Please send me an email to make an appointment.



[This Photo](#) by Unknown Author is licensed under [CC BY-SA](#)

Textbooks and References

- Introduction to Robotics: Mechanics and Control
 - Author: John J. Craig
 - Version: Third or Fourth Edition
 - ISBN: 9780133489835
- Modern Robotics: Mechanics, Planning and Control
 - Authors: Kevin M. Lynch, Frank C. Park
 - Version: First Edition
 - ISBN: 9781107156302



Activities

- Please take some time to read up about the [history of robots](#), and find some [interesting robotic systems](#) in various applications.
- Please share your findings in the Moodle forum.

Thank you for your attention!

Any questions? Please post in Moodle Q&A forum.

References (1)

1. https://commons.wikimedia.org/wiki/File:HONDA_ASIMO.jpg
2. https://commons.wikimedia.org/wiki/File:TOPIO_3.jpg
3. https://commons.wikimedia.org/wiki/File:Atlas_from_boston_dynamics.jpg
4. <https://www.flickr.com/photos/92587836@N04/38795121995>
5. <https://www.flickr.com/photos/jurvetson/32689486>
6. https://commons.wikimedia.org/wiki/File:Jessiko_Robot_Fish_Yeosu2012.jpg
7. <https://commons.wikimedia.org/wiki/File:Roomba3g.jpg>
8. http://i.dailymail.co.uk/i/pix/2014/11/26/23846A8F00000578-0-image-33_1417002924952.jpg
9. [https://commons.wikimedia.org/wiki/File:Care-O-Bot_grasping_an_object_on_the_table_\(5117071459\).jpg](https://commons.wikimedia.org/wiki/File:Care-O-Bot_grasping_an_object_on_the_table_(5117071459).jpg)
10. https://commons.wikimedia.org/wiki/File:Cmglee_Cambridge_Science_Festival_2015_da_Vinci.jpg
11. https://en.wikipedia.org/wiki/Automated_guided_vehicle#/media/File:Forklift_AGV_with_Straddle_courtesy_of_Egemin_Automation_Inc..jpg
12. <https://qz.com/709541/amazon-is-just-beginning-to-use-robots-in-its-warehouses-and-theyre-already-making-a-huge-difference/>

References (2)

13. <https://www.pikrepo.com/fwvfn/two-yellow-plush-toys>
14. https://commons.wikimedia.org/wiki/File:NAO_waving.JPG
15. https://commons.wikimedia.org/wiki/File:Big_Dog_robot_-_D60_Symposium_-_Defense_Advanced_Research_Projects_Agency_-_DSC05557.jpg
16. <https://www.hoa.africom.mil/story/18225/robot-becomes-eyes-and-hands-of-service-members>
17. https://commons.wikimedia.org/wiki/File:Float_Glass_Unloading.jpg
18. http://www.kuka-robotics.com/NR/ronlyres/15C0E6DD-A56A-413C-B16B-812175F7131A/0/Slider_youBot.jpg
19. <https://en.wikipedia.org/wiki/File:Robotworx-arc-welding-robots.jpg>
20. <http://blog.robotiq.com/some-robot-manufacturers-offer-force-control-packages>
21. <http://new.abb.com/products/robotics/application-equipment-and-accessories/integrated-force-control>
22. https://commons.wikimedia.org/wiki/File:Automation_of_foundry_with_robot.jpg
23. https://commons.wikimedia.org/wiki/File:KUKA_Industrial_Robot_KR10_SCARA.jpg
24. https://commons.wikimedia.org/wiki/File:Delta_x_robot.jpg

References (3)

- 25. https://commons.wikimedia.org/wiki/File:KUKA_robot_for_flat_glas_handling.jpg
- 26. <https://www.flickr.com/photos/30478819@N08/36437210460>
- 27. <https://en.wikipedia.org/wiki/File:Robotworx-welding-robot.jpg>
- 28. https://en.wikipedia.org/wiki/Fleming%27s_right-hand_rule